

University of Michigan

Surveys

Micaela Ciambrone, Layilah Campbell, Seorin Jang, Dongsub Kim, Shenghao Mei

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March 10, 2026

Executive Summary

Interviews with USDA professionals identified significant user frustrations regarding search friction and inconsistent regulatory language. We designed a quantitative survey to measure the prevalence of these issues, providing the USDA with scalable data to prioritize updates. The instrument targets findability and interpretive confidence across four sections: Background, Findability, Interpretive Confidence, and Help and Support. All rating scales use 5 point verbally labeled, construct specific formats to minimize bias and ensure high data quality. This survey bridges the gap between qualitative insights and the large scale data needed for portal development.

Introduction

The ACIR portal provides regulatory guidance for importing plant products into the United States, covering over 100,000 country commodity combinations for importers, brokers, and government officials. Earlier interviews with six professionals highlighted recurring pain points, including difficulty locating information via search filters and low confidence in inconsistent regulatory language. While those interviews provided rich context, the prevalence of these issues remains unknown. This survey aims to measure the severity of these pain points at scale, specifically targeting findability and interpretive confidence. Given that the portal is a restricted system, sampling will be conducted internally by the client to maintain security and user anonymity while ensuring a representative sample of its professional user base.

Methods

Research Question and Justification

The survey addresses the following research question: To what extent can ACIR Portal users quickly find the correct requirement information for their shipment, and how confident are they that they interpreted it correctly? This question was informed by our prior interviews, which revealed search autocomplete friction, filter confusion, inconsistent regulatory language, and contradictory classification data between portal files. Findability captures whether users can locate information efficiently; interpretive confidence captures whether they trust their understanding of it once found.

Survey Instrument Design

The survey contains nine questions in four sections: Background (2), Findability (4), Interpretive Confidence (2), and Help and Support (1). All rating scales use 5-point verbally labeled, construct-specific formats rather than numeric scales, which research on satisficing shows are ambiguous and reduce reliability (Krosnick, 1999; Vannette, n.d.). This also avoids the acquiescence bias associated with generic agree-disagree formats.

Target Population and Sampling

The target population for this survey consists of specialized ACIR portal users, including internal APHIS/PPQ staff, CBP officers, and external private stakeholders like customs brokers. We

recommend the client employ **stratified random sampling** to ensure that each of these distinct groups is proportionally represented, preventing the data from being skewed by one specific user type. This approach is necessary because the findability and interpretive challenges faced by a regulatory officer often differ significantly from those of a private importer.

Findings

We distributed our survey to 3 individuals at the USDA via Google Forms. These participants included a technical information specialist, a system manager, and assistant director. We checked for framing bias in questions and whether response options were complete and exhaustive. We selected these individuals as our pilot population because our survey is intended for users of the ACIR portal and these individuals are familiar with the inner workings of the ACIR portal system and how different user groups such as importers, CBP officers, and APHIS or PPQ staff might utilize it.

The pilot group provided high-level feedback such as clarifying multi-select questions to explicitly state that respondents can select multiple options. They also urged us to update terminology, suggesting we replace “Contact APHIS” with “Contact the ACIR Help Desk” in Q6 to better reflect the support channel users would contact. They asked that we clarify the scope of Q8 assessing the clarity of import requirements on the ACIR portal by specifying whether the question refers to the organization/structure of CIR documents and portal display or the language of the regulatory requirements themselves. This distinction is important because regulatory language comes directly from government regulations and cannot be changed while the presentation of information is amendable. A pilot tester also suggested adding the ACIR Help Desk and ACIR Office Hours to Q9, which asks users which help resources they have used.

Looking at the results, we observed moderate search friction with users indicating a range of ease in finding commodity import requirements and using search filters. The results also suggest that search results are sometimes confusing, irrelevant, or inconsistent. Another point raised was that when search fails, users pivot to contacting APHIS inspectors or referring to colleagues rather than using self-service tools. The most urgent finding was the confidence gap in interpreting requirements on the ACIR portal. While 2 users indicated they are very confident in interpreting results, 1 reported being moderately confident, leading us to question whether usability improvements could close this gap.

Based on these findings and group discussions, we identified how our survey could provide insights into the ACIR portal as well as limitations of survey research that make other study methods necessary. Our survey provides quantitative insights into how users view the intuitiveness and navigability of the portal. These insights help form a prioritization matrix to identify pain points for qualitative study procedures. However, the survey is limited to existing users, constraining its scope. It also does not provide in-depth insight into experiences beyond responses to Likert-scale or multi-select questions. These gaps are addressed through other procedures such as semi-structured interviews and usability tests, with the survey providing quantitative measures to complement qualitative insights.

Changes to Survey Instrument

Based on pilot feedback, we made four changes to the survey instrument. First, we added explicit instructions to Q6 and Q9 stating that respondents may select more than one option, as testers noted that the multi-select format was not immediately clear. Second, we revised the Q6 option "Contact APHIS directly" to "Contact the ACIR Help Desk" to better reflect the specific support channel users would actually reach out to rather than the broader agency. Third, we revised Q8 from "How clearly are import requirements written on the ACIR portal?" to "How clearly are import requirements organized and displayed within the ACIR portal?" Pilot testers pointed out that the regulatory language itself comes directly from federal regulations and cannot be modified, while the organization, structure, and display of that information on the portal can be. This revision focuses the question on elements the ACIR team has the ability to change. Fourth, we added two help resources to Q9 that were not in the original draft: ACIR Help Desk and ACIR Office Hours. These are active support channels that the ACIR team offers, and their inclusion allows us to measure awareness of direct support options alongside self-service resources.

Deployment Plan

We recommend a cross channel digital deployment managed internally by the USDA to reach APHIS staff, CBP officers, and trade stakeholders. Because privacy protocols restrict our access to internal directories, the client should distribute the survey via targeted email listservs to registered ACIR users. We also suggest a periodic portal intercept, a non intrusive banner on the ACIR dashboard, to capture feedback while the user experience is top of mind. The survey should remain open for two weeks to ensure adequate data collection. All materials must state that responses are anonymous and used solely for usability improvements to maintain participant trust.

References

- Krosnick, J. A. (1999). Survey research. *Annual Review of Psychology*, 50(1), 537-567.
- Vannette, D. L. (n.d.). *The Qualtrics handbook of question design*. Qualtrics.

Appendix

Appendix A: Final Survey Instrument

Research Question

To what extent can ACIR Portal users quickly find the correct requirement information for their shipment, and how confident are they that they interpreted it correctly?

Target Population

Existing and new ACIR portal users, including importers, customs brokers, CBP officers, and internal APHIS/PPQ staff.

Survey Questions

Background

Q1. What is your role in relation to agricultural imports?

Single select

- a) Importer or customs broker
- b) CBP officer
- c) APHIS/PPQ staff
- d) Other (please specify): _____

Q2. What device do you most often use to access the ACIR portal?

Single select

- a) Desktop or laptop computer
- b) Tablet
- c) Smartphone

Findability

Q3. When you search for a specific commodity's import requirements, how easy or difficult is it to find what you need?

5-point scale

Very easy

Somewhat easy

Neither easy nor difficult

Somewhat difficult

Very difficult

Q4. How well do the search filters help you narrow results to the specific information you need?

5-point scale

Extremely well

Very well
Moderately well
Slightly well
Not well at all

Q5. How often do you encounter search results that are confusing, irrelevant, or inconsistent?

5-point scale

Very often
Often
Sometimes
Rarely
Never

Q6. When you cannot find what you need through the main search, what do you typically do?
(You may select more than one option.)

Select all that apply

- a) Try different search terms or filters
- b) Browse the taxonomy page
- c) Check the Help page
- d) Watch a video or guide
- e) Contact a colleague or supervisor
- f) Contact the ACIR Help Desk
- g) Use a different system (e.g., e-file)
- h) Other (please specify): ____

Interpretive Confidence

Q7. After finding a requirement on the ACIR portal, how confident are you that you have interpreted it correctly?

5-point scale

Extremely confident / Very confident / Moderately confident / Slightly confident / Not at all confident

Q8. How clearly are import requirements organized and displayed within the ACIR portal?

5-point scale

Extremely clearly / Very clearly / Moderately clearly / Slightly clearly / Not at all clearly

Help and Support

Q9. Have you used any of the following help resources on the ACIR portal? (You may select more than one option.)

Select all that apply

- a) Help page
- b) Videos and Guides Hub
- c) In-page labels or tooltips

- d) ACIR Help Desk
- e) ACIR Office Hours
- f) None of the above